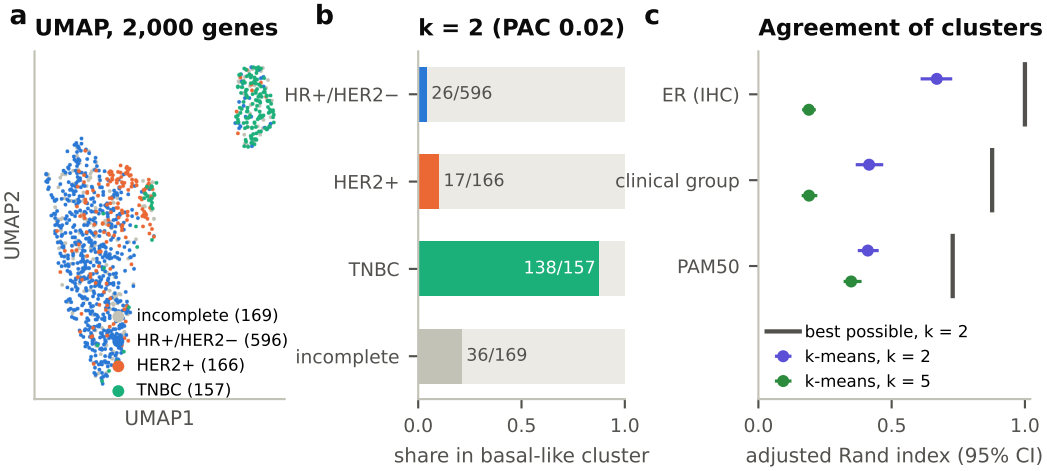

a
UMAP, 2,000 genes
UMAP2
UMAP1
incomplete (169)
HR+/HER2− (596)
HER2+ (166)
TNBC (157)
b
k = 2 (PAC 0.02)
HR+/HER2−
HER2+
TNBC
incomplete
26/596
17/166
138/157
36/169
0.0
0.5
1.0
share in basal-like cluster
c
Agreement of clusters
ER (IHC)
clinical group
PAM50
best possible, k = 2
k-means, k = 2
k-means, k = 5
0.0
0.5
1.0
adjusted Rand index (95% CI)